Akoizo Original Practice - Form 2 - Answer key

Division B | Scrambler | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) How much energy does a 2.00 kg mass release falling 0.50 m?

A. About 19.6 J

B. About 4.9 J

C. About 9.8 J

D. About 1.0 J

Correct: C

2. (2 points) When is the Target Distance announced?

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

A. After impound and before the first time slot

B. Weeks before the tournament

C. During the run

D. After both runs

Correct: A

3. (2 points) If arrival speed doubles with the same stopping distance, how does deceleration change?

A. It quadruples

B. It doubles

C. It halves

D. No change

Correct: A

4. (2 points) Why add O-rings to wheel rims?

Unit: Drivetrains: turning a drop into distance.
Worked-case data: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?
Separate investigation observations:
Run 1: String wound in one neat layer: 8.42 m.
Run 2: String piled in three layers: 7.61 m.
Same: Mass and drop height.

A. To improve traction and prevent slipping

B. To add mass

C. To slow the vehicle

D. To make wheels bigger

Correct: A

5. (4 points) Explain why accuracy matters more than speed.

Unit: Rules and the low-score system.
Worked-case data: A run stops 12.4 cm from the Target Point in 5.31 s with an Inside Can Distance of 60.0 cm and no violations. Find the Run Score.
Separate investigation observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

answer: Each cm costs 2 points while each second costs 1, and runs last only a few seconds, so stopping closer saves more points. (4 points).

6. (4 points) Why do narrow gaps earn more bonus?

Unit: The Can Bonus and curved paths.
Worked-case data: Target 9.00 m. The team sets a 60 cm gap. Find the bonus, the gap center offset, and the arc radius.
Separate investigation observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

answer: The formula subtracts the gap from 110, so smaller gaps give larger negative bonuses. (4 points).

7. (4 points) Why include skid distance in the setting?

Unit: Automatic stopping.
Worked-case data: Target 9.25 m. Wheels are 7.50 cm in diameter. The vehicle skids 3 cm after braking, and the egg tip sits 10 cm ahead of the point used as the start reference. How many turns?
Separate investigation observations:
Setting 30.0 turns: Stops at 7.21, 7.24, 7.19 m.
Setting 35.0 turns: Stops at 8.40, 8.43, 8.38 m.
Target: 8.00 m.

answer: The vehicle keeps moving briefly after the brake triggers, so ignoring skid makes it overshoot. (4 points).

8. (4 points) Compare on-board and launcher designs.

Unit: Energy from a falling mass.
Worked-case data: A 1.50 kg mass falls 0.60 m. If 40 percent reaches a 0.50 kg vehicle as kinetic energy, what is its top speed?
Separate investigation observations:
2.0 kg mass: Top speed 4.2 m/s; egg cracked on stop in 3 of 5 runs.
1.2 kg mass: Top speed 2.6 m/s; egg safe 5 of 5; time +1.8 s.
Target: 8.00 m in both cases.

answer: On-board: the mass drives the wheels as it drops. Launcher: a separate mass pulls the vehicle with a string, then releases it to coast. (4 points).

9. (4 points) Describe a strategy for the second run after a good first run.

Unit: Calibration and competition day.
Worked-case data: Table: 30 turns -> 7.20 m, 34 turns -> 8.15 m, 38 turns -> 9.10 m. Set for a State target of 8.60 m.
Separate investigation observations:
Run 1: Straight run: 4.2 cm from target, 6.1 s, no cans.
Run 1 score: 100 + 8.4 + 6.1 = 114.5.
Option for Run 2: 60 cm can gap; bonus runs average 9 cm error, 6.5 s.

answer: Try a narrower Can Bonus gap or a small calibration improvement, since the good first run is already banked. (4 points).

10. (4 points) How can teams practice taping quickly?

Unit: Protecting the egg.
Worked-case data: A vehicle arrives at 1.5 m/s and stops in 6 cm. Find the average deceleration in g.
Separate investigation observations:
Design A: Arrives 3.0 m/s, stops in 2 cm; 2 of 5 eggs cracked.
Design B: Arrives 1.5 m/s, stops in 5 cm; 0 of 5 cracked.
Question: Compare decelerations.

answer: Rehearse with the same tape type, keeping the front and rear 1.0 cm clear and the rounded end against the backstop. (4 points).

11. (6 points) Solve this case and explain your reasoning: A vehicle has 0.50 cm axle radius and 4.0 cm wheel radius. The mass pulls 0.75 m of string. About how far can the vehicle be driven?

reasoning: Distance ratio = 4.0 ÷ 0.50 = 8.
Distance about 0.75 m x 8 = 6.0 m.
Too short for a 7.00 m minimum target; increase wheel size or string length. (3 points).

conclusion: About 6 m, so the team must raise the ratio or the pull. (3 points).

12. (2 points) Which vehicle scores better?

Supplied synthetic observations:
Vehicle A: Stops 20 cm away in 3.0 s.
Vehicle B: Stops 6 cm away in 7.5 s.
Both: No cans, no violations.

A. Vehicle B

B. Vehicle A

C. They tie

Correct: A

13. (2 points) What explains the short stops?

Supplied synthetic observations:
Arc runs: Passed between cans 10 of 10.
Stopping: Stopped 18 cm short of the target each time.
Setting: Turns calculated from the straight-line distance.

A. The cans slowed the vehicle

B. The egg was too heavy

C. The arc is longer than the straight-line distance

Correct: C

14. (2 points) Define “Stopping mechanism” in the context of Scrambler.

answer: Automatic, in-vehicle device that halts the vehicle. (2 points).

15. (2 points) Define “Kinetic energy” in the context of Scrambler.

answer: ½mv^2 of the moving vehicle. (2 points).

16. (2 points) Define “Calibration table” in the context of Scrambler.

answer: Settings mapped to stopping distances. (2 points).

17. (2 points) Define “Deceleration” in the context of Scrambler.

answer: Rate of slowing, a = v^2 ÷ (2d). (2 points).

18. (2 points) Define “Bearing” in the context of Scrambler.

answer: Part that supports an axle with low friction. (2 points).